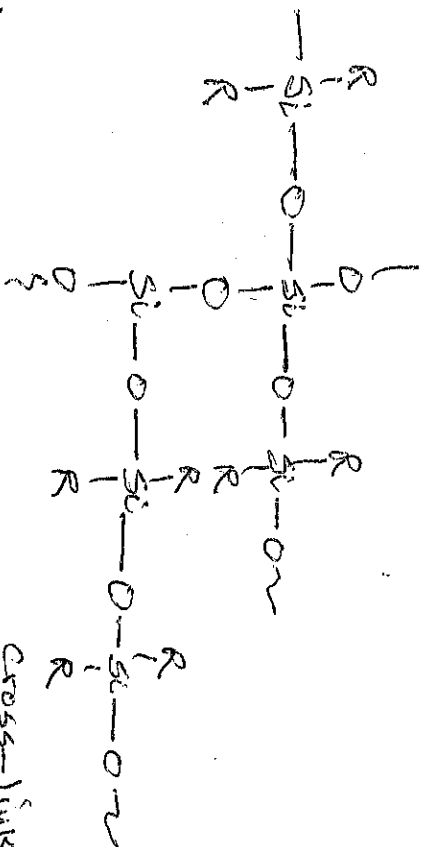




cross-linked silicone

Silicones are the best known of all the synthetic polymers based on inorganic molecular structures. Inorganic polymers gives high ~~valent~~ polymers upon dissolution or melting as shown by their viscosities. Other special characteristics include higher Young modulus, lower failure to strain in comparison to organic polymers.

Silicone polymers are highly stable to many chemical reagents and heat of up to 300°C .

Depending on their degree of polymerization and the chemistry of the attached groups, silicones can be oils, greases, rubber-like substances or resins. They are especially useful as lubricants, hydraulic fluids and electrical insulators. Because of very little changes in their viscosity as temperatures changes, silicones are used as lubricants where there is extreme variation in temperatures.

Paper, wool, silk plus other fabrics are coated with a film of silicone to serve as water-repellent by exposing them to a vapour of trimethyl chlorosilane, $(\text{CH}_3)_3\text{SiCl}$ for a short period of 1-2 sec.

The surface becomes coated with a thin film of $(\text{CH}_3)_3\text{Si}-\text{O}-$ groups as shown.



The most distinguishing property of silicones is their good resistance to oxidation by chemicals, H_2O and stability to heat at high or low temperatures.

Manufacture of Silicones

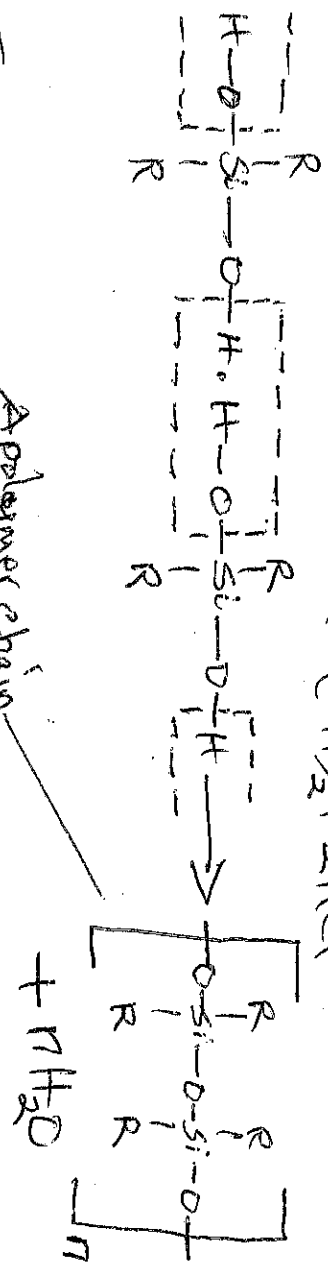
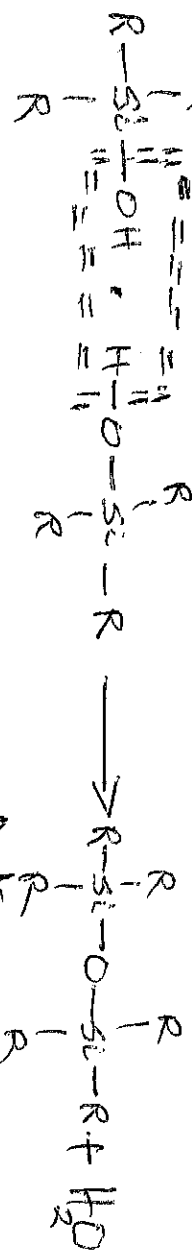
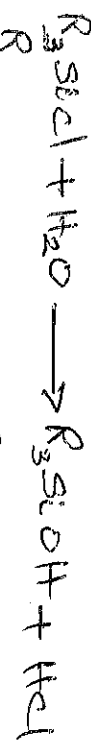
In the laboratory i.e. small scale production of silicones start by reacting a tetrachlorosilane as a Grignard reagent to form alkyl chlorosilanes such as mono, di and trialkyl chlorosilanes such as: $(CH_3)_3SiCl$, $(CH_3)_2SiCl_2$ and $(CH_3)_3SiCl$.

These compounds then undergo hydrolysis and polymerization forming the silicone polymers.

Industrially however, silicones are prepared by the reaction of a chloroalkane or a chlorobenzene with silicon at high temperatures of $300^\circ C$ with a copper catalyst — producing a mixture of alkyl or aryl chlorosilanes.



Upon fractional distillation, the chlorosilanes undergo hydrolysis with a rapid condensation of the hydroxyl groups. The final product depends on the number of hydroxyl groups bonded to the Si-atom originally.



The above chain can be terminated by the addition of a small quantity of the monochlorosilane into the hydrolysis mixture.

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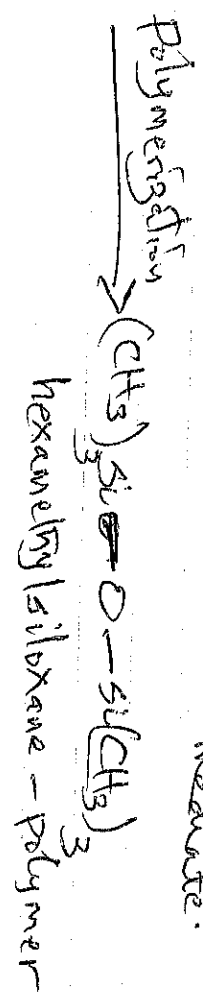
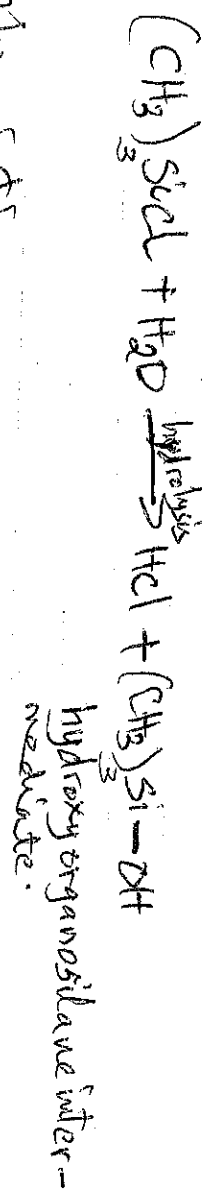
The second reaction is for production of linear and cyclic silicones. A two-dimensional structure or cross-linked silicone is produced by the hydrolysis of a trichlorosilane.

In order to produce polymers of various chain lengths, mixtures of chlorosilanes are blended before the hydrolysis.

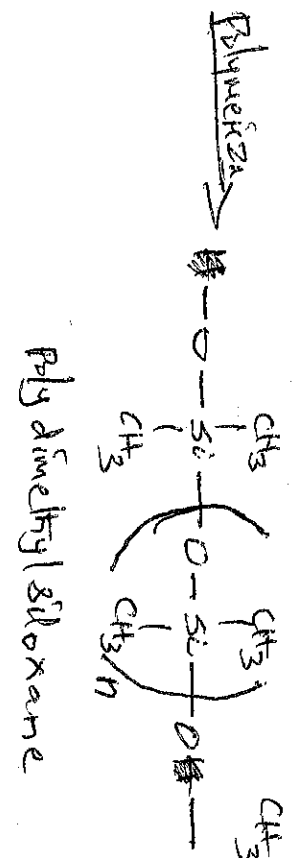
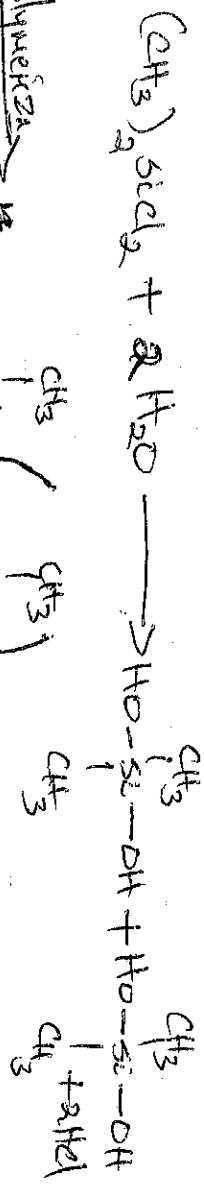
Generally, $R_3Si(OH)_3$ acts as a cross linking agent and R_3SiOH , the chain terminating agent.

Hydrolysis of alkyl chlorosilanes

1, Hydrolysis of trialkyl monochlorosilane.



2, Hydrolysis of a dialkyl dichlorosilane.



3, Hydrolysis of the mixture of trialkyl monochlorosilane and dialkyl dichlorosilane.

5

* Behaviour of Silicones Polymers at high temperatures in absence and presence of air (O_2).

The organic groups of the organosiloxanes shows a wide variation of chemical inertness or reactivity.

The methyl groups in the poly methyl polysiloxane remain attached at temperatures of up to $500^\circ C$, where the siloxane bond or chain depolymerises into volatile cyclic structures and phenyls are equally firmly attached to silicon. However, in the presence of air (O_2), methyl groups begin to be oxidised to aldehydes at $300^\circ C$ or higher whereas ~~methyl~~ ^{phenyl} groups are capable of withstanding temperatures of up to $400^\circ C$ or $450^\circ C$. The higher alkyl groups oxidise at the rate which increases with the chain length so that the higher alkyl silicones have little applications at elevated temperatures. In general, all groups such as halogens, OH, COOH, NH_2 as substituents in the organic part of the molecule decreases the stability of the carbon-silicon bond through their inductive effects on each other.

Properties of Silicones

For instance, silicone rubbers with 87 and 89 high and low temperatures much better than organic rubbers. At temperatures of $150^\circ C$ silicone rubber can be used indefinitely with no change in properties. Silicone polymers/rubbers has excellent resistance to cold temperatures.

Typical organic rubbers have embrittlement point of between $-20^\circ C$ and $-30^\circ C$, whereas silicone rubber is $-60^\circ C$ to $-70^\circ C$. Thus, silicone rubber remains elastic at temperatures where organic rubbers becomes brittle.

Generally, silicone rubbers hardens when heated in air with decreasing elongation as it deteriorates. whereas in sealed conditions it softens as it deteriorates, hence its operating life is shorter in sealed conditions than in air. This is as a result of the ~~slow~~ degradation of the siloxane polymer.

This is controlled by using different curing agents and for post-curing to help prevent softening in hot, sealed conditions.

General Properties of Silicones

Silicones exhibit very useful properties like:

- Low Thermal Conductivity
- Low chemical reactivity
- Low toxicity
- Thermal stability - constancy of properties even at wide temperature ranges of 100 to 350°C.
- Ability to repel water and form seals.
- Non sticky, but adheres very well e.g. on glass.
- Does not support microbiological growth.
- High resistance to oxygen, ozone and UV-light. Leading to widespread use in the construction industry as coatings, fire protection, glazing seals and in the automotive industry as external gaskets, external trim).
- Electrical insulation applications properties.
- High gas permeability - making silicone useful for medical applications in which increased aeration is desired.

Silicone application and uses

~~Silicone~~ have various uses and applications in products and other related areas.

⇒ In the automotive industry silicone grease is used as brake lubricant due to ~~high~~ ^{top} stability ~~at~~ ^{at} high temps and non-multiple layers to prevent spark jumping to adjacent wires. Used as sheet silicone to manufacture gaskets. Also used as sealants and coatings for airbags due to high strength.

⇒ Dry Cleanings; Liquid silicone can be used as dry cleaning solvent alternatively to traditional chlorine-containing solvent such as perchloroethylene.

⇒ In medicine; silicone gel is used in bandages and dressings, breast implants, contact lenses etc.

In ophthalmology silicone has many applications like silicone oil used to replace the vitreous humor following vitrectomy and silicone intracocular lenses following cataract extraction. Personal care; silicone ingredients used in skin care, cosmetic and hair care applications.

INERT GASES

Introduction: The elements in group 8 are the inert (noble gases). ^{note} They six in number viz: helium, neon, argon, krypton, xenon and radon. The names 'inert' or 'noble' were coined to them because they were thought to be unreactive since they all have filled valence shells. The inert gases are stable gases that do not undergo chemical reactions with other elements or gases. ^{note} Therefore, inert means lacking chemical or biological activity or non-reactive. ^{note} There are six primary inert gases located at the far right side of the periodic table and they are as follows:

- Helium (He)
- Neon (Ne)
- Krypton (Kr)
- Argon (Ar)
- Xenon (Xe)
- Radon (Rn)

Electronic Configuration of the inert gases

The element in position eleven in the periodic table is sodium located in the third period. the electronic configuration of sodium is $1s^2 2s^2 2p^6 3s^1$. The first ten electrons of the sodium atom are the inner-shell electrons and the configuration of just those ten electrons is exactly the same as that of the element neon with atomic number 10. this provides the basis for a short hand notation for electronic configurations called the noble gas configuration. Thus, the inert gas electronic configuration of an atom consists of the elemental symbol of the last inert gas prior to that atom, followed by the configuration of the remaining electrons. ^{note} Therefore, for sodium the substitution of [Ne] for the $1s^2 2s^2 2p^6$ part of the configuration hence sodium's inert gas configuration becomes [Ne] $3s^1$. Similarly, the electronic configuration of the inert gases follows the same pattern and are thus given as follows for all the six inert gases except helium:

Helium (He): $1s^2$

Neon (Ne): [Ne] $2s^2 2p^6$

Argon (Ar_{18}): $\text{Ne}[3s^2 3p^6]$

Krypton (Kr_{36}): $\text{Ar}[4s^2 3d^{10} 4p^6]$

Xenon (Xe_{54}): $\text{Kr}[5s^2 4d^{10} 5p^6]$

Radon (Rn_{86}): $\text{Xe}[6s^2 4f^{14} 5d^{10} 6p^6]$

Importance of their electronic configurations

The electronic configurations of the inert gases confer to the gases some important characteristics that relates to their properties and possible reactivity. Some of these are as highlighted below:

1. Because all members of the group 8 elements have 8 electrons in their outermost orbitals with the exception of helium. This makes them to have very stable electronic configuration because of the fully and completely filled valence shells.
2. They have very small atomic radii as compared to other elements and this increases down the group as shells keep on adding.
3. They exhibit higher enthalpies than their counterpart elements.
4. The inert gases have higher ionization energies than any other element $\downarrow$. And this increases down the group as atomic radii increases.

Physical properties of the inert gases

All the inert gases are monatomic, colourless, odourless, at 25°C , and are sparingly soluble in water due to hydrates formation, the table below summarises their physical properties.

Table 1: The inert gases and their physical properties.

Property	He	Ne	Ar	Kr	Xe	Rn
Atomic No.	2	10	18	36	54	86
Electronic configuration	1s ²	2s ² 2p ⁶	3s ² 3p ⁶	4s ² 3d ¹⁰ 4p ⁶	5s ² 4d ¹⁰ 5p ⁶	6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁶
B.P. °C	-269	-246	-186	-152	-107	-62
M.P. °C	-272	-249	-189	-157	-112	-71
Percent by volume in air	5.2x10 ⁻⁴	1.8x10 ⁻³	0.93	1.1x10 ⁻⁴	8.7x10 ⁻⁶	0.00

Compounds of the inert gases

There was a long held belief that the inert gases are unreactive hence the name inert/noble gases. this is because the atoms of these elements have little or no tendencies of accepting extra electrons in their outermost shells which gives them great stability. However, in 1962, compounds of the noble gases have been prepared by Neil Bartlett by exposing xenon to platinum hexafluoride resulting into the reaction below:



From the viewpoint of chemistry, the noble gases may be divided into two groups the relatively reactive Krypton with ionisation energy (IE = 14.0 eV), Xenon (IE = 12.1 eV), and Radon (IE = 10.7 eV) on one side, and the very unreactive Argon (IE = 15.8 eV),

Neon (IE = 21.6 eV), and Helium (IE = 24.6 eV) on the other. Consistent with this classification; Xe, Kr and Rn form compounds that can be isolated in bulk at or near standard temperature and pressure, whereas He, Ne, Ar have been observed using spectroscopic techniques, to form true chemical bonds but only when frozen into a noble gas matrix at temperatures of 40 K or lower, under extremely high pressures with metals.

The heavier noble gases have more electron shells than the lighter ones. Hence, the outermost electrons are subject to a shielding effect from the inner electrons that makes them more easily ionisable, since they are less strongly attracted to the positively charged nucleus. This results in an ionization energy low enough to form stable compounds with the most electronegative elements like fluorine and oxygen. In a similar manner, other compounds of xenon such as (XeF₂, XeF₄, XeF₆, XeO₃, XeO₄, XeOF₄ and KrF₂ (very unstable)) have been prepared. Stable compounds of xenon form when xenon reacts with fluorine gas

forming xenon difluoride, XeF_2 . After heating an excess of the xenon gas with the fluorine gas and then cooling, the compound formed is colourless crystals that are stable at room temperature in a dry atmosphere. Xenon tetrafluoride, XeF_4 and Xenon hexafluoride, XeF_6 are prepared in the same manner with stoichiometric amount of fluorine and an excess of fluorine gas respectively.

Compounds with oxygen are prepared by placing fluorine atoms in the xenon fluoride with oxygen. Radon forms RnF_2 in radiochemical tracer techniques. Unstable compounds of argon form at low temperatures, however, there are no known stable compounds of helium and neon.

Formation of Clathrates and coordination compounds by the inert gases

Clathrate formation: Clathrates are compounds or 'enclosure compounds' or molecules of three dimensional structures that contain cavities two thirds of which is occupied by smaller molecules such as argon, methane, nitrogen, etc.; and which escapes upon heating or dissolving because the cavities are not fully filled. Clathrates are formed by compressing noble gases in water, where it is believed that the water molecule, with a stronger dipole, induces a weak dipole in the noble gas atoms, resulting in a dipole-dipole interaction. Heavier atoms are more influenced than smaller ones, the important condition for the formation of clathrates is that the guest (noble gas-atoms usually in the middle) should be of appropriate size to fit in the cavities of the host crystal lattice; for instance, Ar, Kr, and Xe can form clathrates with crystalline β -quinol, but He and Ne cannot fit because they are too small. For instance, a clathrate is formed when hydroquinone (1,4-dihydroxybenzene) is crystallized from benzene or water under argon at 20 atm, and the formula will be as shown $[\text{C}_6\text{H}_4(\text{OH})_2]_3 \cdot \text{Ar}$. Helium and Neon do not form stable clathrates perhaps because of their too small sizes and as such easily escape. Hence, a method for separating helium and neon from the rest of the noble/inert gases. Clathrates containing the inert gases are sometimes called hydrates and is their terminology in the oil and gas industry.

Coordination compounds formation

Coordination compounds such as $\text{Ar} \cdot \text{BF}_3$ have been postulated to exist at low temperatures, but have never been confirmed. Also, compounds such as $\text{W} \cdot \text{He}_2$ and $\text{Hg} \cdot \text{He}_2$ were reported to have been formed by electron bombardment, but recent research has shown that these are

probably the result of He being adsorbed on the surface of the metal; therefore, these compounds cannot truly be considered chemical compounds. In practice therefore, the inert gases do not form significant coordination compounds because of their poor or near non-reactivity in general.

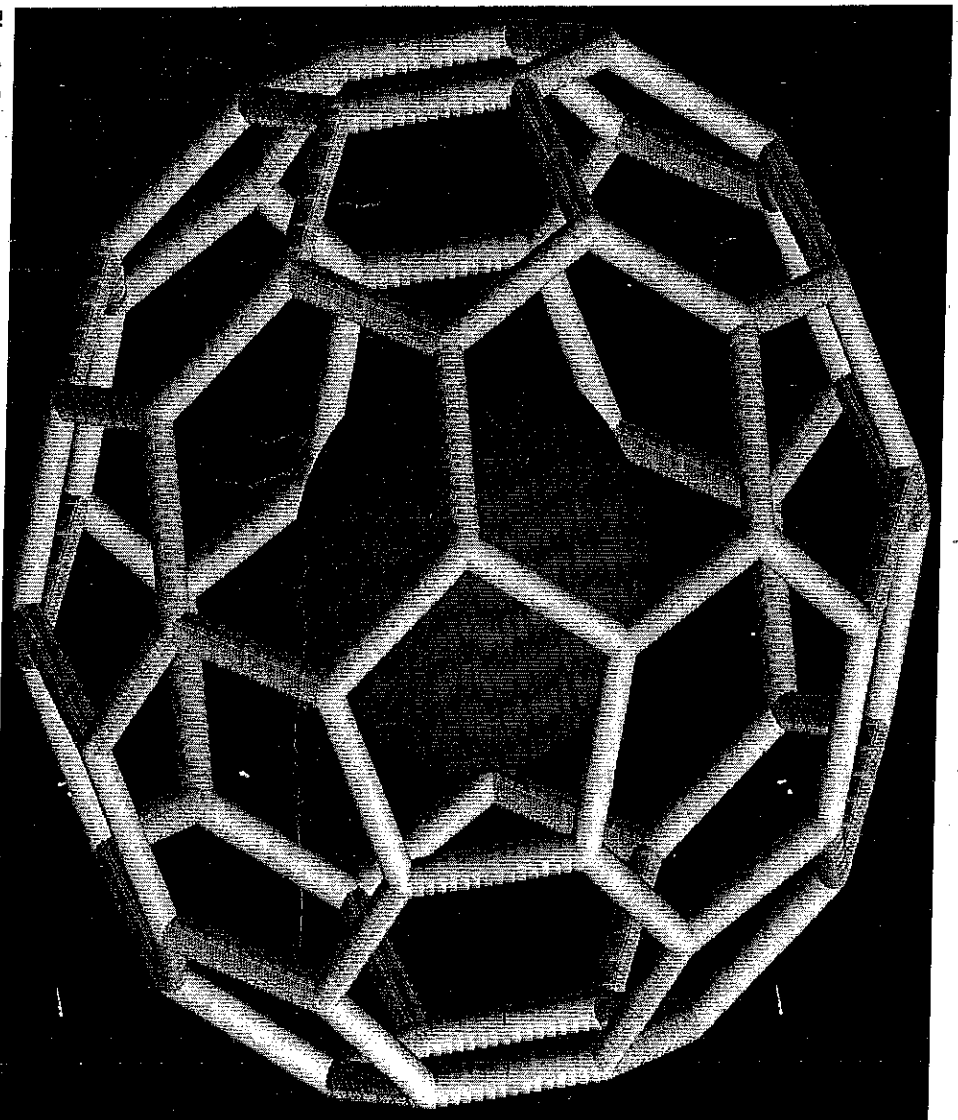
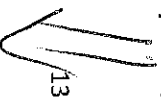


Fig. 1: Structure of an inert-gas atom (He, Ar) caged within a fullerene molecule.

Production of the inert gases

All of the six inert gases except radon have been conveniently produced from fractional distillation of liquid air. Also higher concentrations of helium gas about 10% is found in some natural gases especially those of the USA and Canada. The helium is separated by condensation of the other constituents. Additionally, the inert gases also occur in low concentrations in the atmosphere.

Application/ Use of inert gases



The most identified application of the inert gases or their uses in practice is not very diverse. However, on the individual level the most recent applications of the following gases are outlined.

1. Helium is used for filling balloons as it is lighter than air. And also safer because it does not burn as hydrogen does. Helium at higher pressures is not a narcotic like nitrogen. Thus, mixtures of oxygen and helium are used by deep divers under high pressures. Helium is also important as an inert atmosphere for welding and melting of easily oxidisable metals and for many chemical processes sensitive to air. Helium having a very low boiling point 4.2 K, serves as an important coolant in for the magnets used in NMR imaging in medical diagnostic procedure. Another coolant used is nitrogen gas with boiling point of 77 K and which is significantly cheaper.
2. Neon is used in neon lamps and signs. When an electric spark is passed through a tube containing neon gas at low pressure it generates the familiar red glow of neon. The colour of the light can be changed by mixing argon or mercury vapour with the neon gas or by utilizing glass tubes of a special colour.
3. Argon is used in tungsten electric light bulbs because of its lower heat of conductivity thereby inhibiting the vaporization of the tungsten filament, hence prolonging the life of the bulb.
4. Krypton-xenon flash tubes are used to take high speed photographs.